



Structural basis and affinity improvement for an ATP-binding DNA aptamer

Yan Jiang^{a,b}, Yuchao Zhang^b , Liqi Wan^b , Cheng Cui^a, Pei Guo^{b,1} , Da Han^{b,c,1} , and Weihong Tan^{a,b,c,1}

Affiliations are included on p. 8.

Edited by Dinshaw Patel, Memorial Sloan Kettering Cancer Center, New York, NY; received March 21, 2025; accepted July 10, 2025

DNA aptamers that bind small molecules with high affinity have revolutionized the fields of biosensing and bioimaging. Recently, a DNA aptamer named 1301b has been identified as the most potent DNA aptamer for the binding of adenosine triphosphate (ATP) with a dissociation constant (K_D) of ~ 2.7 μM . However, the structural basis and recognition mechanism remain unclear, hindering further development of this DNA aptamer. In this study, we first design a shortened DNA aptamer namely 1301b_v1 that retains a good affinity for ATP and then determine the tertiary structure of 1:1 1301b_v1-ATP binding complex using solution NMR spectroscopy. The overall complex structure shows an “L” shape architecture with the binding pocket formed by two internal loops. The ATP intercalates into the binding pocket through forming hydrogen bond with G14 and stacking with T8-A28 and G9. We also reveal an adaptive binding mechanism where the DNA aptamer switches from a semifolded state to a stable tertiary structure upon ATP binding. Based on the structure–function relationship, we introduce 2'-O-methyl modification to residues in the central junction and obtain a DNA aptamer named 9/10/16^{OMe} with a K_D of ~ 0.7 μM for the binding of ATP. These results underscore the ability of DNA molecules to form intricate three-dimensional folds with sophisticated functionality, opening up avenues for designing novel DNA-based molecular tools.

DNA aptamer | DNA structure | high-resolution structure | solution NMR

DNA molecules are best known for the canonical double-helical B-form structures, which underpin their essential roles in storing genetic information (1). Advancements in the technique of systematic evolution of ligands by exponential enrichment (SELEX) have revealed a remarkable structural and functional diversity of DNA molecules, represented by DNA aptamers that are capable of forming three-dimensional (3D) structures to confer sophisticated biochemical and biophysical activities (2, 3). In particular, DNA aptamers that can bind small molecules, such as metabolites, drugs, and fluorophores, have found widespread applications in biosensing, bioimaging, and bioanalysis (4–7). Elucidating the structural basis and recognition mechanism of DNA aptamers is critical for post-SELEX optimization and downstream applications (8, 9). However, DNA molecules are generally less adept than RNA molecules at forming tertiary interactions and higher-order structures (10–13), posing a significant barrier to structural characterization. In addition, computational tools such as the AlphaFold have yet to excel in predicting 3D structures of DNA–ligand complexes (14, 15). The number of resolved structures of DNA aptamer–ligand complexes remains far fewer than those of RNA aptamer–ligand complexes (16), limiting our understanding of the structure–function relationship of DNA aptamers. Consequently, studying DNA aptamers with intricate 3D folds and advanced functionalities is propelling to the research field (17, 18).

Adenosine triphosphate (ATP), a central metabolite in cellular energy metabolism, is a key small-molecule target for aptamer development. Notably, two classic ATP aptamers have been reported (*SI Appendix, Fig. S1*), including a 27-nt DNA aptamer with a dissociation constant (K_D) of ~ 6 μM (19) and a 40-nt RNA aptamer with a K_D of ~ 0.7 μM for binding of ATP (20). Their complex structures with adenosine monophosphate (AMP), but not ATP, have been resolved. The DNA aptamer binds two molecules of AMP within two symmetric simple binding pockets (21), whereas the RNA aptamer forms a more delicate S-shaped binding pocket to accommodate one molecule of AMP (22, 23). Despite its lower affinity for ATP, the DNA aptamer is more widely used in biosensing and therapeutic applications due to its higher chemical stability and programmability (24, 25). Recently, a new 40-nt DNA aptamer named 1301b has been reported to bind one molecule of ATP with a K_D of ~ 2.7 μM (Fig. 1A) (26). However, the structural basis and recognition mechanism of this DNA aptamer remain

Significance

DNA aptamers that bind small molecules with high affinity are essential tools in biosensing and bioimaging, yet a poor understanding on their structure–function relationships hinders effective design. Numerous efforts devoted to developing DNA aptamers with high affinity for adenosine triphosphate (ATP), a central metabolite in cellular energy metabolism, encounter challenges. Here, we determine the tertiary structure of a recently discovered DNA aptamer in complex with one molecule of ATP, and elucidate the binding mechanism. Leveraging the structural insights, we engineer a DNA aptamer with a K_D of ~ 0.7 μM for ATP binding. This study demonstrates the ability of DNA to harness structural and functional complexity, paving the way for designing high-performance DNA aptamers toward multiple applications.

Author contributions: P.G., D.H., and W.T. designed research; Y.J. and Y.Z. performed research; Y.J., Y.Z., L.W., and P.G. analyzed data; and Y.J., C.C., P.G., D.H., and W.T. wrote the paper.

The authors declare no competing interest.

This article is a PNAS Direct Submission.

Copyright © 2025 the Author(s). Published by PNAS. This article is distributed under [Creative Commons Attribution-NonCommercial-NoDerivatives License 4.0 \(CC BY-NC-ND\)](#).

¹To whom correspondence may be addressed. Email: guopei@ibmc.ac.cn, dahan@sjtu.edu.cn, or tan@hnu.edu.cn.

This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2506491122/-/DCSupplemental>.

Published August 14, 2025.

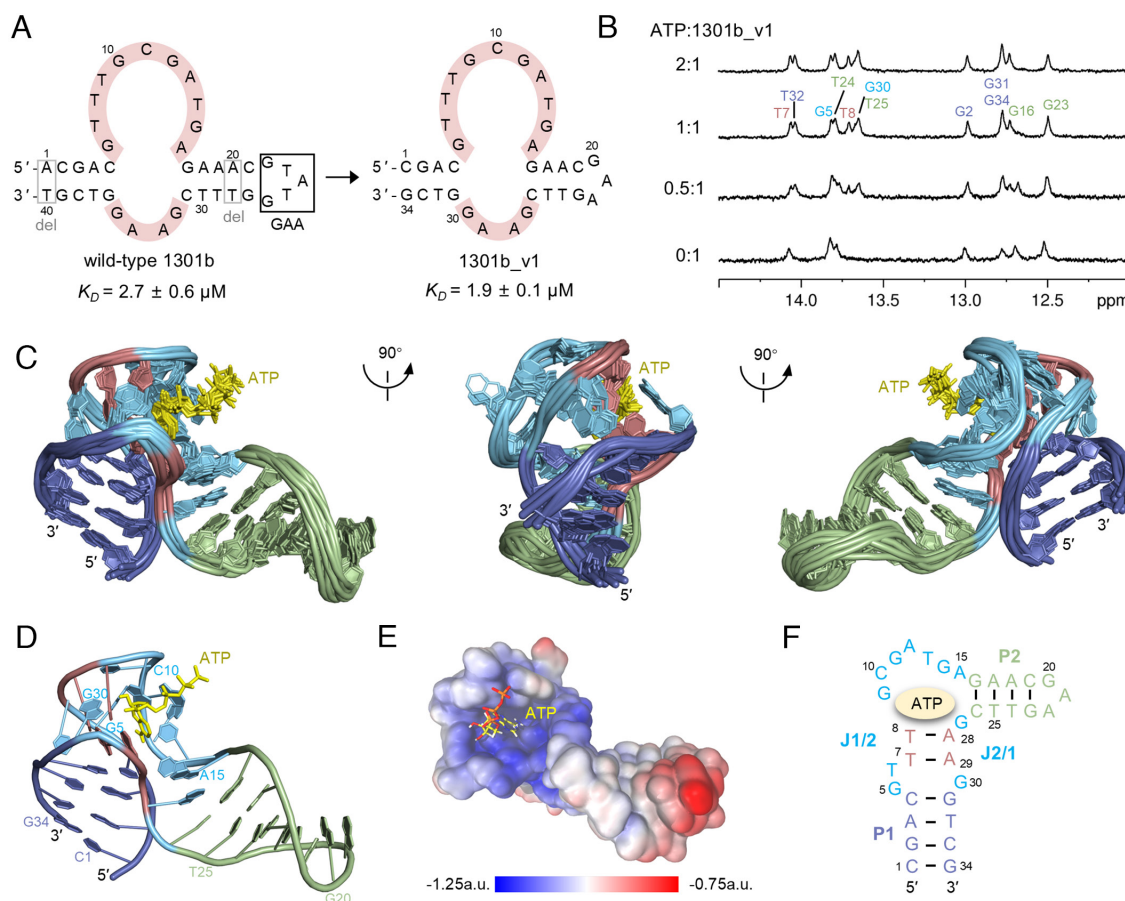


Fig. 1. DNA aptamer modification and 3D structure of 1301b_v1-ATP complex. (A) The secondary structure and K_D values of the wild-type 1301b (Left) and 1301b_v1 (Right). Residues to be deleted and mutated in the wild-type 1301b are indicated by grey and black rectangular boxes, respectively. The two internal loops are shown in pink. (B) 1D ^1H NMR spectra showing G H1 and T H3 signals of 1301b_v1 upon ATP titration. [DNA] = 0.1 mM, [ATP] = 0/0.05/0.1/0.2 mM, [NaPi, pH 7] = 10 mM, [NaCl] = 50 mM, [MgCl₂] = 10 mM, 90% H₂O/10% D₂O, T = 20 °C. (C) The superimposed 10 solution NMR structures of 1301b_v1-ATP complex in different views (PDB ID: 9KTJ). (D) The cartoon mode showing one of the 10 solution NMR structures. (E) The electrostatic potential mapped on the van der Waals surface. (F) Schematic showing the secondary structure of 1301b_v1-ATP complex. Black lines denote Watson-Crick base pairs.

unclear, which hinders further improvement and application of this DNA aptamer.

Here, we determine the tertiary structure of the 1:1 DNA aptamer-ATP complex and elucidate the recognition mechanism using solution NMR spectroscopy. The complex structure adopts an “L” shape architecture, with a delicate binding pocket formed by two internal loops to favorably enclose ATP. In addition, we reveal that the ATP recognition involves an adaptive binding mechanism, where divalent Mg²⁺ ions facilitate the formation of a semifolded DNA structure which is further stabilized upon ATP binding. Finally, we use 2'-O-methyl modifications to engineer a DNA aptamer, namely 9/10/16^{OMe}, with a K_D of ~0.7 μM for ATP binding, representing the first DNA aptamer with a submicromolar K_D for ATP binding. These findings extend the structural paradigm of DNA and underscore the untapped potential of DNA as important functional biomolecules.

Results

DNA Aptamer Modification and Validation. To facilitate solution NMR structural study, a shorter aptamer maintaining the affinity is desired. In the wild-type 1301b, the two internal loops (nucleotides #6-16 and #32-35) are suggested to constitute the essential binding pocket (26). Therefore, we retain the two internal loops, delete A1-T40 and A20-T28 base pairs that are located far from the internal loops, and further compensate the reduced structural stability by replacing the original GTATG loop

(nucleotides #22-26) with a thermodynamically ultrastable GAA loop (27) (Fig. 1A). The shortened aptamer is 34-nt and named as 1301b_v1, which is then subjected for binding assay by isothermal titration calorimetry (ITC). Compared to the wild-type 1301b with a K_D of $2.7 \pm 0.6 \mu\text{M}$, 1301b_v1 exhibits a smaller K_D of $1.9 \pm 0.1 \mu\text{M}$ for ATP binding (Fig. 1A and *SI Appendix, Fig. S2*).

We next investigate the binding of 1301b_v1 and ATP by 1D ^1H NMR spectroscopy (Fig. 1B). Specifically, we monitor the guanine imino proton (G H1) and thymine imino proton (T H3) signals at 12.5 to 14.5 ppm that provide clues for the formation of base pairs. The NMR buffer contains 10 mM NaPi (pH 7), 50 mM NaCl, and 10 mM MgCl₂ unless otherwise specified. The free 1301b_v1 shows only seven G H1 and T H3 signals, suggesting that the free DNA aptamer does not form a well-folded tertiary structure in solution. Adding 1 equivalent of ATP leads to the appearance of 12 intense and well-resolved G H1 and T H3 signals from the binding complex. This suggests that the ATP recognition involves an adaptive binding, where ATP induces a conformational change of 1301b_v1 in the binding pocket to form a stable tertiary structure. The NMR spectra of 1301b_v1 at 1 and 2 equivalents of ATP are nearly identical, suggesting that the aptamer is saturated by 1 equivalent of ATP and agreeing with the binding stoichiometry determined by ITC (*SI Appendix, Fig. S2*). We also acquire 1D ^1H NMR spectra for the wild-type 1301b upon adding ATP (*SI Appendix, Fig. S3*) and verify that the wild-type 1301b and 1301b_v1 behave similarly. Overall, a shortened aptamer with slightly

improved affinity is successfully designed for high-resolution NMR structural study.

3D Structure of 1301b_v1-ATP Binding Complex. For the 1:1 1301b_v1-ATP binding complex, we assign proton resonances of 1301b_v1 from 2D nuclear Overhauser effect spectroscopy (NOESY) and correlation spectroscopy following the established NMR protocols (28). In addition, the resonances of some G H1 signals are confirmed with the aid of site-specific ^{15}N -isotope labeling on guanine residues (*SI Appendix, Fig. S4*). A total of 10 Watson–Crick base pairs, including C1·G34, G2·C33, A3·T32, C4·G31, T7·A29, T8·A28, G16·C26, A17·T25, A18·T24 and C19·G23, are identified based on their corresponding G H1–C H41/H42 NOEs or T H3–A H2 NOEs (*SI Appendix, Figs. S5 and S6*). The high-resolution structure of 1301b_v1–ATP complex is determined by restrained molecular dynamics (rMD) simulations with NMR experimental restraints. Ten structures with the lowest total energies are selected as the final representative ensemble (Fig. 1C) and their NMR refinement statistics are shown in *SI Appendix, Table S1*.

The 1301b_v1–ATP complex structure is shown in cartoon, surface, and schematic representations in Fig. 1D–F. The complex structure adopts an “L” shape architecture, comprising two paired regions including P1 (nucleotides #1–4 and 31–34) and P2 (nucleotides #16–26), as well as two internal loops or junctions including J1/2 (nucleotides #5–15) and J2/1 (nucleotides #27–30). In the two internal loops, T7·A29 and T8·A28 form Watson–Crick base pairs. For the GAA loop in P2, G20·A22 adopts a sheared base pair (*SI Appendix, Fig. S7*). We also perform small-angle X-ray scattering (SAXS) experiment for the 1301b_v1–ATP complex in 10 mM MgCl_2 (*SI Appendix, Fig. S8*). The SAXS data result in a linear fit in the Guinier region, demonstrating that the sample is monodisperse. The dimensionless Kratky plot is consistent with a well-folded globular structure (29). The $P(r)$ distribution is well behaved, displaying a zero value at $P(0)$ and ending smoothly at maximum dimension (D_{max}). The D_{max} (55 Å) and radius of gyration (R_g) (16.2 Å) determined by SAXS (*SI Appendix, Table S2*) are close to the D_{max} (54 Å) and R_g (15.6 Å) measured from 10

solution NMR structures. Overall, the 1301b_v1–ATP complex adopts a compact tertiary structure in solution.

Stabilization Interactions of the Internal Loops. In the internal loops, one notable feature is that T7·A29 and T8·A28 form two Watson–Crick base pairs (Fig. 2A and B). This is supported by strong NOEs of T7 H3–A29 H2 and T8 H3–A28 H2 (*SI Appendix, Fig. S6*), as well as the chemical shifts of T7 H3 at 14.09 ppm and T8 H3 at 13.72 ppm that agree with those of T·A Watson–Crick base pairs (30). T7·A29 and T8·A28 base pairs stack well, with T7 O2 and T8 O4' forming additional hydrogen bonds with G30 H21 and G30 H1, respectively (Fig. 2B and C). G5 extrudes into the loop turn formed by G9 to T13, using its H21, H22, and H1 to form hydrogen bonds with C10 O3', C10 O2, and G11 OP2, respectively (Fig. 2D). In the loop turn, T13 uses its O2, H3 and O4 to form hydrogen bonds with G11 H22, G11 N3, and G9 H1/H21, respectively (Fig. 2E). The hydrogen bonds shown in Fig. 2C–E are predominantly observed in the 10 solution NMR structures (*SI Appendix, Figs. S9–S11*). In addition, A15, G27, and C26 form continuous base–base stackings (Fig. 2F). Taken together, extensive hydrogen bonds and base–base stackings stabilize the two internal loops to form a central junction for ATP binding.

Architecture of the ATP-Binding Pocket and Recognition Mechanism. The ATP-binding pocket is located at the interface of T8·A28 and G9 to G14 in the central junction. The nucleobase of ATP intercalates into the binding pocket, leaving its sugar and phosphates outward from the binding pocket (Fig. 3A). Specifically, the nucleobase of ATP stacks between T8·A28 and G9 (Fig. 3B and C), which agrees with the more upfield aromatic H2 and H8 resonances of ATP in the binding complex than those of free ATP due to ring current effects, as well as the NOEs between ATP and T8·A28 (*SI Appendix, Fig. S12*). To assess the importance of these stacking interactions mediated by T8·A28 and G9, we perform site-directed mutation assays by substituting i) T8·A28 to A·T, G·C, or C·G, and ii) G9 to A, C, or T. All these mutations disrupt the binding (Fig. 3E and

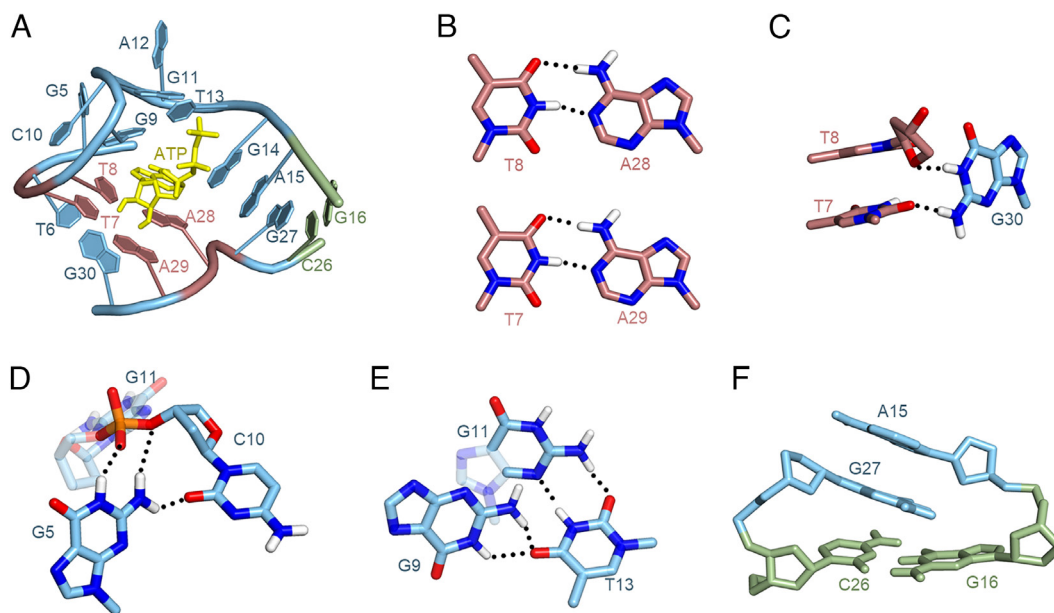


Fig. 2. Stabilizing interactions of the central junction. (A) An overall view of the central junction structure of 1301b_v1–ATP complex. (B) Two Watson–Crick base pairs formed by T7·A29 and T8·A28 in the internal loops. (C) T7 O2 and T8 O4' form hydrogen bonds with G30 H21 and G30 H1, respectively. (D) G5 H21, G5 H22, and G5 H1 form hydrogen bonds with C10 O3', C10 O2, and G11 OP2, respectively. (E) T13 O2, T13 H3, and T13 O4 form hydrogen bonds with G11 H22, G11 N3, and G9 H1/H21, respectively. (F) A15 stacks with G27, and G27 further stacks on G16·C26.

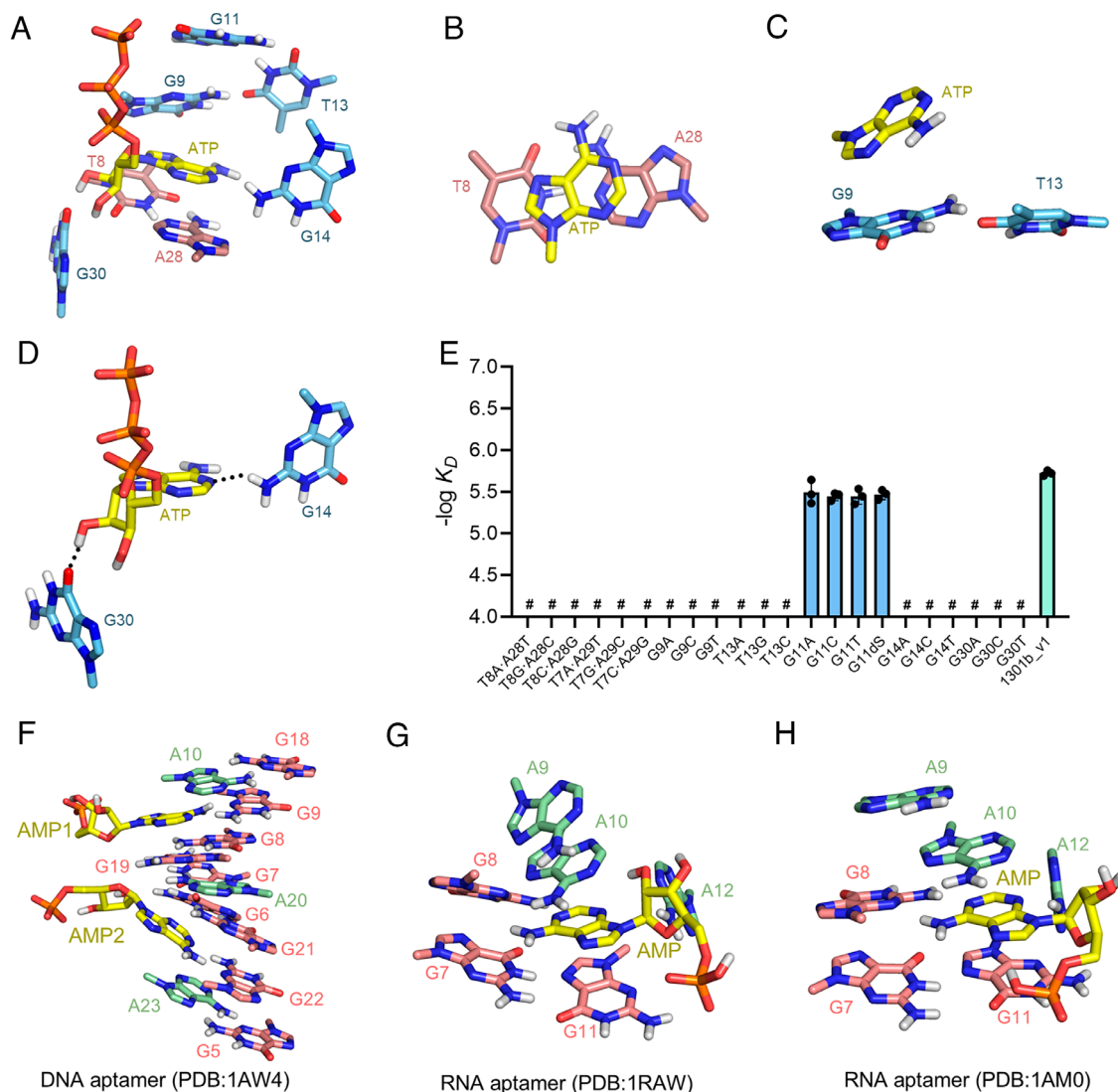


Fig. 3. Architecture of the binding pocket. (A) An overall view of the binding pocket structure. (B and C) The nucleobase of ATP stacks with T8·A28 base pair and G9. (D) N1 of ATP forms a hydrogen bond with G14 H21/H22, and HO2' of ATP forms a hydrogen bond with G30 O6. The aptamer and ATP are shown in stick mode. (E) ITC analysis of ATP binding of 1301b_v1 mutants. “#” means no binding. Data represent mean $\pm$ SD from three independent ITC measurements. The original ITC data are shown in *SI Appendix, Figs. S13–S15 and S17*. (F) The binding pocket structure of the classic DNA aptamer in complex with two molecules of AMP (PDB ID: 1AW4) (21). (G and H) The binding pocket structures of the 36-nt RNA aptamer–AMP complex (PDB ID: 1RAW) (22) and the 40-nt RNA aptamer–AMP complex (PDB ID: 1AM0) (23).

SI Appendix, Figs. S13 and S14). While the T7·A29 base pair does not directly interact with ATP, it stacks with T8·A28 and appears to be a crucial structural element. Substitution of T7·A29 to A·T, G·C, or C·G also results in no binding (Fig. 3E and *SI Appendix, Fig. S13*). For T13 that helps to stabilize G9 by forming hydrogen bonds (Fig. 2E), its mutation abolishes the binding (Fig. 3E and *SI Appendix, Fig. S14*). For G11 that does not directly interact with ATP, its phosphate and nucleobase form hydrogen bonds with G5 and T13, respectively (Fig. 2D and E). Substitution of G11 to A/C/T or an abasic site (G11dS) retains the binding with larger K_D values of 3.3 to 3.6 μ M, but deletion of G11 results in no binding (Fig. 3E and *SI Appendix, Fig. S15*). These suggest that the phosphodiester backbone of the #11 residue plays an important role in the binding, whereas the hydrogen bond formed between G11 nucleobase and T13 is not crucial to the binding. Apart from base–base stackings, ATP is further stabilized by several hydrogen bonds, including ATP N1...G14 H21/H22, and ATP HO2'...G30 O6/T8 O2 (Fig. 3D and *SI Appendix, Fig. S16*). Accordingly, substitution of G14 or G30 to A/C/T results in no binding (Fig. 3E and *SI Appendix, Fig. S17*). In addition, a higher

K_D of 3.4 ± 0.2 μ M for 1301b_v1 binding to deoxyadenosine triphosphate (dATP), which lacks the 2'-OH hydroxyl group compared to ATP, also suggests that the hydroxyl group of ATP contributes to the binding (*SI Appendix, Fig. S18*).

We compare the binding pocket structures of 1301b_v1-ATP with the reported classic DNA aptamer–AMP and RNA aptamer–AMP complexes. The classic DNA aptamer forms a relatively simple stem-loop structure, where two AMP-binding sites locate in the helical stem (*SI Appendix, Fig. S19A*) (21). Each AMP forms two hydrogen bonds with the opposite guanine, and intercalates between neighboring G·A and G·G mismatches (Fig. 3F). For the classic RNA aptamer, a 36-nt and 40-nt RNA in complex with AMP have been determined by two research groups, respectively, with similar findings (22, 23). Using the 36-nt one to illustrate, the two stems of RNA are oriented almost orthogonally to each other, resulting in an “L” shape architecture with a binding pocket composed of two internal loops (*SI Appendix, Fig. S19B*). The intercalation of AMP between A10 and G11 constitutes a GAAA loop, with AMP serving as the fourth loop residue by forming hydrogen bonds with its opposite G8 (PDB ID: 1RAW,

Fig. 3*G*). The GAAA loop is a well-characterized ultrastable structural element in RNA (31). The 40-nt RNA aptamer–AMP complex shows a similar binding pocket and recognition mechanism (PDB ID: 1AM0, Fig. 3*H*). This comparative analysis suggests that despite differences in the detailed binding pocket structures of 1301b_v1, classic DNA and RNA aptamers, the recognition of ATP and its derivatives is mainly achieved by forming hydrogen bonds with the opposite guanine and assisted by stackings with neighboring bases. In 1301b_v1–ATP complex, an intriguing feature is that T7·A29 and T8·A28 from the two internal loops form Watson–Crick base pairs, assisting the intercalation of ATP to form hydrogen bond with G14. We also demonstrate that 1301b_v1 does not bind GTP, CTP, TTP, or UTP (*SI Appendix, Fig. S20*), which is consistent with the behavior of wild-type 1301b (26). In the 1301b_v1–ATP complex, ATP N1 forms a hydrogen bond with G14 H21/H22 (Fig. 3*D*). We construct four putative models by replacing ATP with GTP/CTP/TTP/UTP and find that GTP H1 has steric hindrance, instead of forming hydrogen bond, with G14 H21/H22, whereas the nucleobase of CTP/TTP/UTP is too distant from G14 to form hydrogen bond with G14 (*SI Appendix, Fig. S21*).

Mg²⁺ Ions Induce Folding of DNA Aptamer and Stabilize ATP Binding. It has been reported that divalent Mg²⁺ ions are required for the binding of ATP by the 1301b DNA aptamer (26), but the underlying mechanism is unclear. To investigate this, we

perform rMD simulations for the 1301b_v1–ATP complex in explicit solvents supplemented with Mg²⁺ ions. Analysis of the Mg²⁺ trajectories shows that Mg²⁺ ions predominantly locate in the binding pocket where electrostatic potentials are relatively more negative (Figs. 1*E* and 4*A*). Specifically, Mg²⁺ ions locate most frequently between the phosphate groups of G14/A15 and triphosphate group of ATP, and between the phosphate groups of vicinal DNA strands in the central junction (Fig. 4*A*). These suggest that Mg²⁺ ions stabilize the binding complex by neutralizing negatively charged phosphate groups.

To gain more detailed insights, we acquire 1D ¹H NMR spectra to monitor how Mg²⁺ ions facilitate the folding of free DNA aptamer and the binding of ATP (Fig. 4*B*). In the absence of Mg²⁺, the free 1301b_v1 shows G16 H1, G23 H1, T24 H3, and T25 H3 signals that are suggestive of forming only four Watson–Crick base pairs in P2. Adding 1/5/10/20 mM Mg²⁺ leads to the appearance of G2 H1, G31 H1, and T32 H3 signals that indicate the formation of three Watson–Crick base pairs in P1. The NMR spectra of free 1301b_v1 look similar in 10 and 20 mM Mg²⁺, which agrees with the ITC result that a plateau of affinity is achieved in 10 and 20 mM Mg²⁺ (Fig. 4*C* and *SI Appendix, Fig. S22*). In 0 mM Mg²⁺, NMR and ITC results show no binding between 1301b_v1 and ATP (Fig. 4*B* and *C*). Besides, the SAXS data show that 1301b_v1 is not well folded in the absence of Mg²⁺ (*SI Appendix, Fig. S8* and Table S2). Taken together, these results suggest that Mg²⁺ ions facilitate 1301b_v1 to form a semifolded

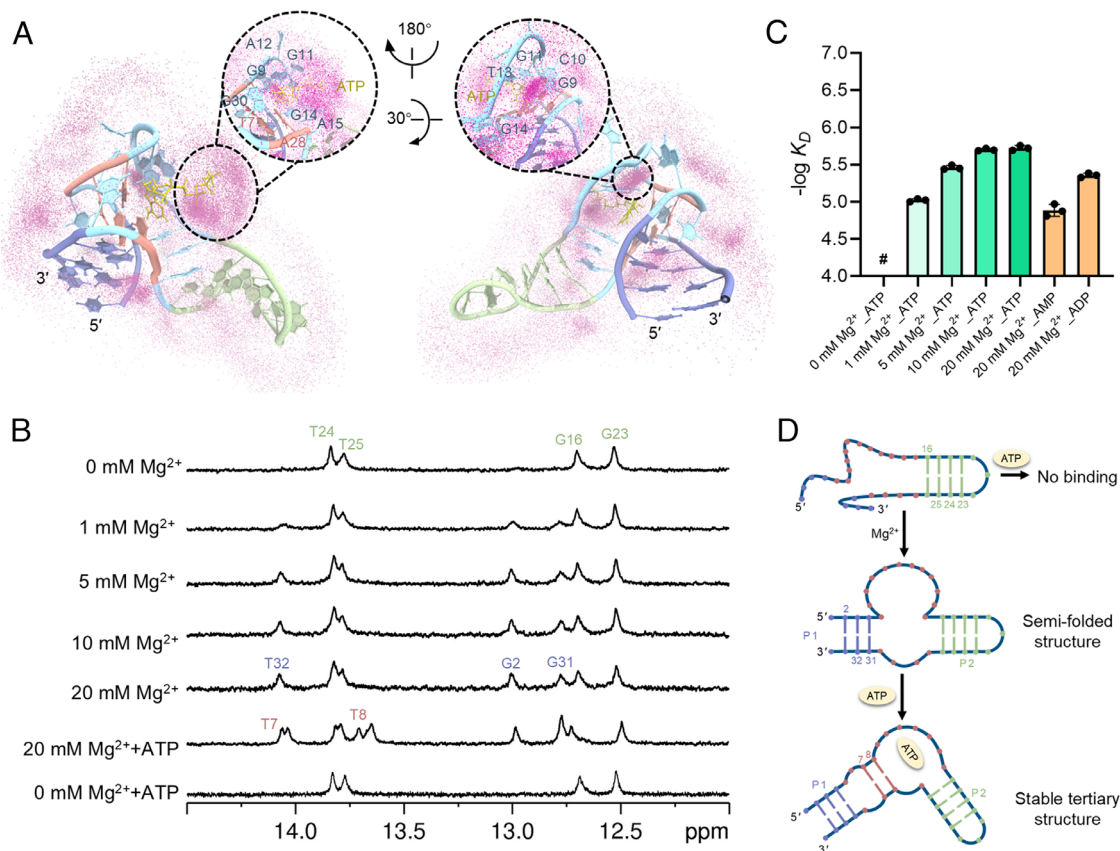


Fig. 4. Mg²⁺ ions stabilize the 1301b_v1–ATP complex. (A) Trajectories of Mg²⁺ ions during rMD simulations for 1301b_v1–ATP complex. Enlarged views in the dotted black circles show the two most frequent locations of Mg²⁺ ions. The Mg²⁺ ions are represented by magenta dots. (B) 1D ¹H NMR spectra show G H1 and T H3 signals of 1301b_v1 with 0/1/5/10/20 mM Mg²⁺ in the absence of ATP (Top five), and with 20/0 mM Mg²⁺ in the presence of 0.1 mM ATP (Bottom two). [DNA] = 0.1 mM, [NaPi, pH 7] = 10 mM, [NaCl] = 50 mM, 90% H₂O/10% D₂O, T = 20 °C. (C) ITC analysis of ATP binding of 1301b_v1 in 0/1/5/10/20 mM Mg²⁺, and AMP and ADP binding of 1301b_v1 in 20 mM Mg²⁺. “#” means no binding. Data represent mean ± SD from three independent ITC measurements. The original ITC data are shown in *SI Appendix, Figs. S22, S24, and S25*. (D) In the absence of Mg²⁺ ions, 1301b_v1 does not form a stable P1 stem and cannot bind ATP (Top). Mg²⁺ ions facilitate the formation of base pairs in P1, leading to a semifolded structure (Middle), which adaptively switches to a stable tertiary structure upon ATP binding (Bottom).

structure and further stabilize the binding complex by neutralizing the negatively charged phosphate groups of DNA and ATP (Fig. 4D).

As for the triphosphate group, the α , β , and γ ^{31}P resonances of ATP in complex with 1301b_v1 show an overall broadening compared to those of free ATP (*SI Appendix*, Fig. S23), suggesting that the phosphates of ATP undergo moderate local conformational dynamics in the NMR time scale. To investigate the effect of phosphate(s) of ATP on binding, we determine the K_D of 1301b_v1 for adenosine diphosphate (ADP) and AMP in 0/1/5/20 mM Mg^{2+} (*SI Appendix*, Figs. S24 and S25). The binding to ADP and AMP is improved when increasing Mg^{2+} concentration but is no better than that to ATP under the same Mg^{2+} concentration. For example, the K_D of 1301b_v1 to ATP, ADP, and AMP are 1.9 ± 0.1 , 4.4 ± 0.3 , and 13.2 ± 2.4 μM , respectively, in 20 mM Mg^{2+} . This agrees with the previous finding that the binding of wild-type 1301b follows the order of ATP > ADP > AMP (26). It is suggested that increasing the number of phosphate units facilitates Mg^{2+} -mediated bridging interactions (32, 33). We further perform MD simulations on 1301b_v1-ADP and 1301b_v1-AMP complexes in explicit solvents supplemented with Mg^{2+} and find lower frequencies of Mg^{2+} ions distributed around the phosphate(s) of ADP/AMP compared to ATP (Fig. 4A and *SI Appendix*, Fig. S26).

Structure-Guided Functional Optimization. DNA aptamers with high affinity for small molecules are highly desired toward downstream applications such as biosensing and bioimaging (34), but achieving a submicromolar K_D is still challenging (26, 35). Prior to this work, the wild-type 1301b represents a DNA aptamer

with the highest affinity for ATP (K_D of ~ 2.7 μM), which is nearly the lowest ATP concentration limit in SELEX and difficult to be further improved (26). In addition, the K_D of ~ 2.7 μM requires a Mg^{2+} concentration as high as 20 mM that is far beyond the physiological Mg^{2+} concentration of 0.5 to 1 mM (36), thus limiting its biological applications.

Based on the structure–function relationship established above, we seek to engineer new aptamers with improved affinity for ATP and reduced reliance on Mg^{2+} . For residues that constitute the binding pocket, their nucleobases are generally intolerant to mutation (Fig. 3E). Therefore, our focus is directed to sugar modification. 2'-O-methylation (2'-O-Me) is a commonly used modification to improve the performance of functional nucleic acids such as aptamers and ribozymes (37, 38). It has been reported that 2'-O-Me can stabilize alternative conformations of bulges and loops (39). For 1301b_v1, its central junction exhibits a higher degree of conformational dynamics and requires more Mg^{2+} for ATP binding than the helical stem region (Fig. 4A and D). We wonder whether 2'-O-Me can help to stabilize the central junction and improve ATP binding. We first perform 2'-O-Me modification on individual nucleotide of #8–16 and #26–28 (Fig. 5A). G9^{OMe}, C10^{OMe}, G14^{OMe}, and G16^{OMe} improve the binding, G11^{OMe}, A12^{OMe}, and A28^{OMe} weaken the binding, while T8^{OMe}, T13^{OMe}, A15^{OMe}, C26^{OMe}, and G27^{OMe} result in no binding (Fig. 5B and *SI Appendix*, Figs. S27–S29).

We then perform 2'-O-Me modifications simultaneously on G9, C10, G14, and G16, namely 9/10/14/16^{OMe}, but fail to improve the binding ($K_D = 3.8 \pm 0.4$ μM in 20 mM Mg^{2+}). We further design four mutants by triple 2'-O-Me modifications, including 9/10/14^{OMe}, 9/10/16^{OMe}, 9/14/16^{OMe}, and 10/14/16^{OMe}, among

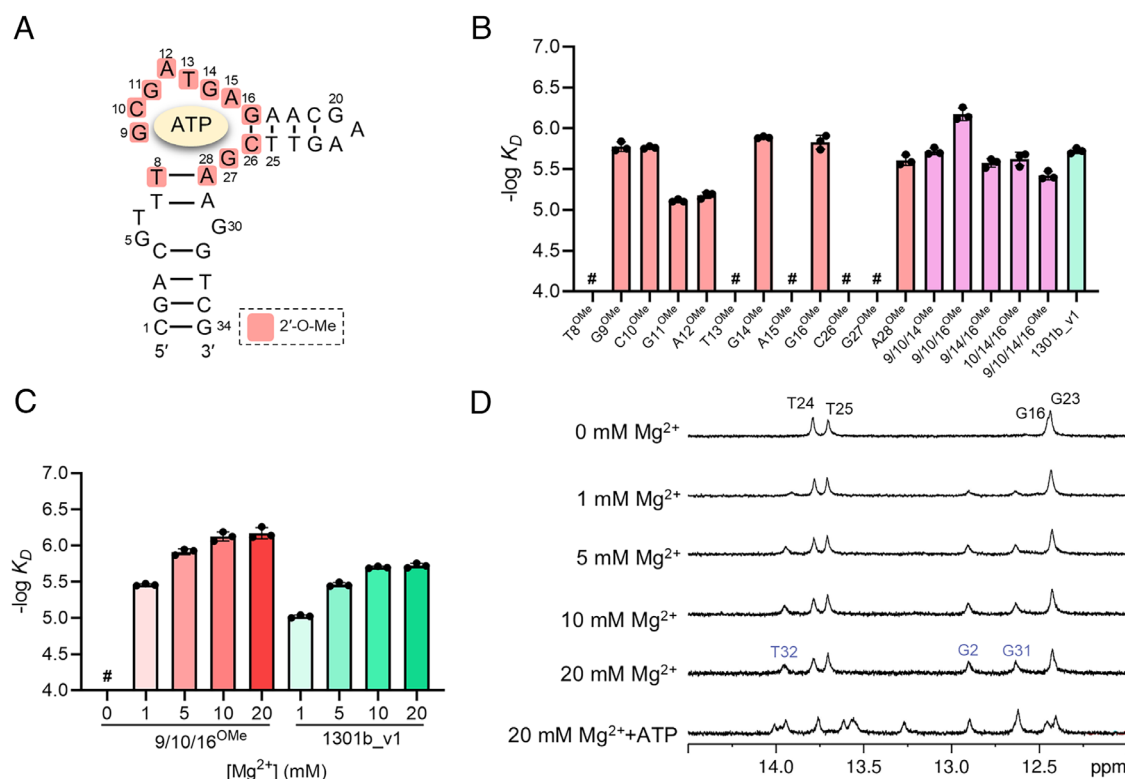


Fig. 5. Structure-guided functional optimization. (A) Schematic of 1301b_v1-ATP secondary structure shows 2'-O-Me modification on residues in the central junction (red). (B) ITC analysis of ATP binding of 1301b_v1 variants in 20 mM Mg^{2+} . “#” means no binding. Data represent mean $\pm$ SD from three independent ITC measurements. The original ITC data are shown in *SI Appendix*, Figs. S27–S29. (C) ITC analysis of ATP binding of 9/10/16^{OMe} and 1301b_v1 under different Mg^{2+} concentrations. “#” means no binding. Data represent mean $\pm$ SD from three independent ITC measurements. The original ITC data are shown in *SI Appendix*, Fig. S30. The data of 1301b_v1 in (C) is identical to those of 1301b_v1 in Fig. 4C. (D) 1D ^1H NMR spectra show G H1 and T H3 signals of 9/10/16^{OMe} in 0/1/5/10/20 mM Mg^{2+} without adding ATP (Top five) and 20 mM Mg^{2+} with one equivalent of ATP (Bottom). [DNA] = 0.1 mM, [NaPi, pH 7] = 10 mM, [NaCl] = 50 mM, 90% H_2O /10% D_2O , T = 20 $^\circ\text{C}$.

which 9/10/16^{OMe} shows the smallest K_D of $0.7 \pm 0.1 \mu\text{M}$ in 20 mM Mg^{2+} . In addition, 9/10/16^{OMe} also exhibits a lower reliance on Mg^{2+} concentration. The K_D of 9/10/16^{OMe} is $3.5 \pm 0.1 \mu\text{M}$ in 1 mM Mg^{2+} (Fig. 5C and *SI Appendix, Fig. S30*), which is much smaller than that of 1301b_v1 in 1 mM Mg^{2+} ($K_D = 9.5 \pm 0.4 \mu\text{M}$) and comparable to that of 1301b_v1 in 5 mM Mg^{2+} ($K_D = 3.5 \pm 0.2 \mu\text{M}$). Analysis of the binding enthalpy and entropy components suggests that compared to 1301b_v1, the improved affinity and lower Mg^{2+} reliance of 9/10/16^{OMe} are likely to be attributed to favorable entropic contributions (40) (*SI Appendix, Tables S3–S5*). Moreover, we demonstrate that 9/10/16^{OMe} does not bind GTP, CTP, TTP, or UTP, retaining the binding specificity for ATP (*SI Appendix, Fig. S31*). We further validate that 9/10/16^{OMe} recognizes ATP through an adaptive binding mechanism (Fig. 5D). Overall, 9/10/16^{OMe} represents the first DNA aptamer with a submicromolar K_D for ATP, underscoring the accuracy and efficacy of our 3D structure-guided functional optimization.

Discussion

In this work, we elucidate the structural basis for a newly identified DNA aptamer that binds one molecule of ATP. The complex structure adopts an “L” shape architecture, where the two internal loops form a central junction to encapsulate ATP (Fig. 1C–E). The recognition of ATP is achieved by forming a hydrogen bond with G14 and facilitated by stackings with T8-A28 and G9 (Fig. 3A–D). In addition, we explore the role of Mg^{2+} ions in DNA folding and ATP binding, revealing that Mg^{2+} ions facilitate the free DNA aptamer to form a semifolded state and further stabilize ATP binding by neutralizing the negatively charged phosphate groups of DNA and ATP (Fig. 4). Furthermore, we modify sugar moieties of several key residues in the central junction, significantly improving the K_D to $\sim 0.7 \mu\text{M}$ (Fig. 5). This work brings the following insights to the field of functional DNA molecules.

Exploring the ability of DNA molecules to form higher-order structures remains an interesting topic (17, 18). Tertiary interactions are commonly observed in RNA molecules and they contribute to the formation of higher-order RNA structures. By contrast, DNA molecules typically lack the capability of forming intricate 3D folds. Recent studies have revealed the four-way junction fold of a 53-nt Lettuce DNA aptamer and the three-way junction fold of a 41-nt *sgc8c* DNA aptamer, expanding the structural diversity of DNA molecules. The classic DNA and RNA aptamers for binding of ATP were developed three decades ago, with optimal K_D of $\sim 6 \mu\text{M}$ and $0.7 \mu\text{M}$, respectively (19, 20). From the structural point of view (Fig. 3F–H and *SI Appendix, Fig. S19*), the classic DNA aptamer forms a relatively simple stem-loop structure with two binding sites in the helical stem (21), whereas the RNA aptamer forms a tertiary structure with a more delicate binding pocket composed of two internal loops (22, 23). The binding of AMP constitutes a GAAA loop, which is a well-characterized ultrastable structural element in RNA (31) and may account for the superior affinity of the RNA aptamer. For the recently developed 1301b DNA aptamer, its affinity for ATP is higher than the classic DNA aptamer. We determine the tertiary structure of the 1301b_v1–ATP complex, revealing a delicate binding pocket composed of two internal loops. In particular, T7-A29 and T8-A28 from the internal loops form Watson–Crick base pairs to stabilize the binding pocket (Fig. 2A–C). The recognition of ATP is achieved through forming a hydrogen bond with G14 and stacking with T8-A28 and G9 (Fig. 3B–D). The structural findings underscore the ability of DNA molecules to form intricate 3D architectures that exert sophisticated functions.

It has been well documented that structure-switching aptamers have applications in biosensing and bioimaging, such as monitoring dynamical changes of targets in complex biological environment (41, 42). However, structure-switching DNA aptamers with intricate 3D folds are rarely found. For example, the Lettuce DNA aptamer, despite its delicate 3D fold, does not undergo a structural change upon ligand binding (17). By taking advantage of solution NMR spectroscopy, we elucidate the adaptive binding mechanism of 1301b_v1. The DNA aptamer forms a semifolded structure in the absence of ATP and switches to a stable tertiary structure upon ATP binding (Fig. 4D), revealing the dynamic recognition ability of 1301b_v1. In addition, our designed 9/10/16^{OMe} also preserves the dynamic recognition ability (Fig. 5D). Recent studies have designed structure-switching aptamers combined with high-throughput sequencing to quantify metabolites including ATP (34), and utilized the ATP-binding DNA aptamer to improve the biostability of other DNA aptamers in tumor microenvironment where ATP concentration is high (25). Considering the low Mg^{2+} concentration in extracellular fluids (43), we anticipate that our optimized 9/10/16^{OMe} with structure-switching behavior, higher affinity and lower Mg^{2+} reliance for ATP binding can further advance such applications.

In sum, we report a 1:1 DNA aptamer–ATP complex structure and elucidate the ATP recognition mechanism. The 3D structural information and site-directed mutation assay collectively establish structure–function relationships of the DNA aptamer. We engineer an aptamer with a submicromolar K_D for ATP binding, which represents the currently best ATP-binding DNA aptamer. Overall, our work extends the structural paradigm of DNA molecules and provides principles for the development of functional DNA molecules.

Materials and Methods

Sample Preparation. DNA oligonucleotides were purchased from Sangon Biotech (Shanghai, China) and were of high-performance liquid chromatography (HPLC) purification grade. The 6% ^{15}N isotopically labeled DNA oligonucleotides were synthesized on a K&A H8 synthesizer using the 2′-deoxyguanosine phosphoramidite (98% ^{15}N and 98% ^{13}C) purchased from Cambridge Isotope Laboratories (USA) and purified by HPLC. All the oligonucleotides were further purified in our laboratory using diethylaminoethyl sephacel anion exchange column and centrifugal desalting. DNA samples were quantified using a NanoDrop microvolume spectrophotometer. The DNA sequences used in this study are shown in *SI Appendix, Table S6*.

NMR Spectroscopy. NMR samples were prepared to contain $\sim 0.1 \text{ mM}$ DNA (for 1D experiments) to $\sim 0.8 \text{ mM}$ DNA (for 2D experiments) in 10 mM sodium phosphate (NaPi, pH 7) buffer with 50 mM NaCl and 10 mM MgCl_2 unless otherwise specified. Excitation sculpting and presaturation water suppression methods were applied for 90% $\text{H}_2\text{O}/10\% \text{D}_2\text{O}$ and 99.96% D_2O samples, respectively. 2D NMR experiments, including NOESY, total correlation spectroscopy (TOCSY), double quantum filtered-correlation spectroscopy, and ^1H – ^{15}N heteronuclear single quantum correlation (HSQC) were acquired using a Bruker AVANCE 600 MHz spectrometer. NOESY was acquired with a mixing time of 200 ms. TOCSY was acquired with a mixing time of 75 ms. ^1H – ^{15}N HSQC was acquired with a $^1J_{\text{N-H}} = 90 \text{ Hz}$. NMR spectra were acquired at 20°C unless otherwise specified. NMR data were analyzed using TopSpin 3.6.5 software.

Structural Calculations. The DNA structure was calculated by rMD simulations on GROMACS 2021.7 (44) using NOE-derived distance restraints in time averaging form (45), as well as backbone restraints, hydrogen bond distance, and planarity restraints for Watson–Crick base pairs in 1–34, 2–33, 3–32, 4–31, 7–29, 8–28, 16–26, 17–25, 18–24, and 19–23. The complex of DNA aptamer and ATP was simulated using the CHARMM36 force field (46) in 10 mM Na_2HPO_4 , 50 mM NaCl, and 10 mM MgCl_2 with modified water model (47). The restraints were

employed during energy minimization (restraints in permanent form), heating, equilibration, annealing, and production. Langevin integrator (48) and stochastic cell rescaling barostat (49) were used to control temperature and pressure in all molecular dynamics steps. All bonds with hydrogen atoms were constrained with LINCS algorithms (50).

The structure calculation was divided into two phases, which were conformation sampling and structure refinement. In conformation sampling phase, the system was linearly heated from 0 to 298.15 K in 0.8 ns under 1 atm pressure after energy minimization and then, continually equilibrated for 1.2 ns. Then, 250 times periodic annealing (heating the system to 363.15 K in 100 ps and keeping the temperature for 400 ps, then cooling to 298.15 K in 100 ps and keeping 1.4 ns) were conducted on the system to sampling its accessible conformations as more as possible (planarity restraints were removed and force constants of distance restraints were set to 1/5 of its previous value in this process). The trajectories in 298.15 K were extracted and combined for analysis (1.4 ns * 250 = 350 ns, 17,500 frames). All structures were clustered, and then each cluster was sorted by restraint violation. The 20 clusters with the least restraint violations were selected, and their centroid structures were set as the initial structures.

In the structure refinement phase, rMD simulations of each initial structure were performed independently by annealing and sampling for five times. In brief, each replica was heated to 353.15 K in 100 ps, kept for 400 ps, cooled to 298.15 K in 100 ps, and continually equilibrated for 400 ps. The trajectories of the next 20 ns simulation of each replica were combined for analysis (20 ns * 5 replicas = 100 ns, totally 2 μ s and 200,000 frames for 20 initial structures). The 20 most frequent structures for each trajectory were singled out, respectively, and then 20 * 20 = 400 structural clusters were obtained. We analyzed the 33 largest clusters (contents more than 300 frames). Then, the 30 structures with the least restraint violations were selected and subjected to restrained energy minimization using Ambertools 23 (51). The 10 lowest energy structures serve as the final representative ensemble. 3D structural figures were prepared using PyMOL unless otherwise stated.

The spatial distribution of Mg^{2+} ions was calculated by rMD simulations using the similar condition as in structural calculations. The NOE-derived distance restraints, bond distance, and planarity restraints for Watson-Crick base pairs were still employed. The concentration of $MgCl_2$ was set to 30 mM for efficient sampling the spatial distribution. After energy minimization, a 200-ps heat process from 0 K to 298.15 K, a 300-ps equilibrium and a 500-ns production simulation were run to obtained the trajectory. The trajectories (500 ns, 125,000 frames) of aptamer-ATP complex were aligned using least squares fit and the Mg^{2+} ions are shown as points each 130 ps (68 ps in enlarged view). Only the Mg^{2+} within a distance of 5 Å to the complex are shown. The 3D structural figure of aptamer-ATP complex in Fig. 4A was prepared using VMD (52). The same procedure was performed on aptamer-ADP and aptamer-AMP complexes.

Electrostatic potential Φ was calculated from atomic partial charges in CHARMM36 force field using formula below by Multiwfn 3.8(dev) program (53):

$$\Phi(\mathbf{r}) = \sum_i \frac{q_i}{|\mathbf{r} - \mathbf{r}_i|},$$

where q_i and $\mathbf{r}_i$ are atomic partial charge and coordinate of atom i , respectively.

ITC Experiment. All the ITC experiments were performed on a Nano-ITC instrument and a Micro-Cal PEAQ-ITC microcalorimeter. Before each experiment, the cell chamber and syringe were cleaned with Milli-Q water carefully. The ITC buffer was prepared following the SELEX buffer to contain 50 mM Tris (pH 7.6), 500 mM NaCl, and 20 mM $MgCl_2$, unless otherwise specified. DNA aptamers and target molecules were dissolved in the ITC buffer. Then, 0.3 mL of 10 μ M aptamer was loaded into the cell chamber and 50 μ L of 200 μ M, 400 μ M, or 1 mM target was loaded into the syringe. The titration experiment was carried out at 25 °C. Each time, 2 μ L target molecule was injected. The time between each injection was 120 s. All ITC data files were analyzed with Nano-Analyze software and Micro-Cal PEAQ-ITC analysis software, and the titration curves were plotted by Origin software. The binding and thermodynamic parameters determined by ITC including the K_D , enthalpic contribution (ΔH), and entropic contribution ($-\Delta S$) are summarized in *SI Appendix, Tables S3–S5*.

SAXS Experiment. The SAXS sample contains 0.1 to 0.2 mM purified DNA in 10 mM Tris-HCl (pH 7.5), 50 mM NaCl, and 0/10 mM $MgCl_2$. The SAXS data were collected at beamline BL19U2 of the Shanghai Synchrotron Radiation Facility, with a radiation wavelength of 1.03 Å (12.0 keV). Each sample was measured 20 times, and the sample measurements were adjusted by subtracting the scattering from the buffer alone. Data were analyzed using the software package RAW (<https://sourceforge.net/projects/bioxtasraw/>). The scattering images were averaged and subtracted from the buffer-scattering images. Then, using the indirect Fourier transform method, the R_g was estimated. The distribution function $P(r)$ was calculated from the parameter as D_{max} .

Data, Materials, and Software Availability. The atomic coordinates of the 1301b_v1-ATP complex are deposited to the Protein Data Bank under the accession code: **9KTJ** (54). All other data are included in the manuscript and/or *SI Appendix*.

ACKNOWLEDGMENTS. This work is supported by the National Key Research and Development Program of China (2021YFA0909400), National Natural Science Foundation of China (22225402, 32341017, 22374132), Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang Province (2024R01005), Natural Science Foundation of Zhejiang Province (QKHM25B0501), and State Key Laboratory of Fine Chemicals, Dalian University of Technology (KF2101). We thank the staff members of BL19U2 beamline at the National Facility for Protein Science in Shanghai (<https://cstr.cn/31129.02.NFPS.BL19U2>) for providing technical support and assistance in data collection and analysis. We acknowledge the support from the Scientific Experiment Center from Hangzhou Institute of Medicine Chinese Academy of Sciences.

Author affiliations: ^aMolecular Science and Biomedicine Laboratory, State Key Laboratory of Chemo and Biosensing, College of Chemistry and Chemical Engineering, College of Biology, Aptamer Engineering Center of Hunan Province, Hunan University, Changsha, Hunan 410082, China; ^bZhejiang Key Laboratory of Functional Nucleic Acids for Basic and Clinical Application, Hangzhou Institute of Medicine, Chinese Academy of Sciences, Hangzhou, Zhejiang 310022, China; and ^cInstitute of Molecular Medicine, Renji Hospital, School of Medicine, Shanghai Jiao Tong University, Shanghai 200240, China

- J. D. Watson, F. H. C. Crick, Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid. *Nature* **171**, 737–738 (1953).
- M. R. Dunn, R. M. Jimenez, J. C. Chaput, Analysis of aptamer discovery and technology. *Nat. Rev. Chem.* **1**, 0076 (2017).
- X. Wu *et al.*, Efficient strategy to discover DNA aptamers against low abundance cell surface proteins in scarce samples. *J. Am. Chem. Soc.* **146**, 26667–26675 (2024).
- G. Xu *et al.*, Structure-guided post-SELEX optimization of an ochratoxin A aptamer. *Nucleic Acids Res.* **47**, 5963–5972 (2019).
- Y. Wu *et al.*, Genetically encoded fluorogenic DNA aptamers for imaging metabolite in living cells. *J. Am. Chem. Soc.* **147**, 1529–1541 (2025).
- P.-J. J. Huang, J. Liu, A DNA aptamer for theophylline with ultrahigh selectivity reminiscent of the classic RNA aptamer. *ACS Chem. Biol.* **17**, 2121–2129 (2022).
- C. D. Flynn *et al.*, Biomolecular sensors for advanced physiological monitoring. *Nat. Rev. Bioeng.* **1**, 560–575 (2023).
- A. Brown, J. Brill, R. Amini, C. Nurmi, Y. Li, Development of better aptamers: Structured library approaches, selection methods, and chemical modifications. *Angew. Chem. Int. Ed. Engl.* **63**, e202318665 (2024).
- G. Xu *et al.*, Structural insights into the mechanism of high-affinity binding of ochratoxin A by a DNA aptamer. *J. Am. Chem. Soc.* **144**, 7731–7740 (2022).
- M. L. Ken *et al.*, RNA conformational propensities determine cellular activity. *Nature* **617**, 835–841 (2023).
- L. R. Ganser, M. L. Kelly, D. Herschlag, H. M. Al-Hashimi, The roles of structural dynamics in the cellular functions of RNAs. *Nat. Rev. Mol. Cell Biol.* **20**, 474–489 (2019).
- K. Huang *et al.*, Structure-based investigation of fluorogenic Pepper aptamer. *Nat. Chem. Biol.* **17**, 1289–1295 (2021).
- R. J. Trachman *et al.*, Structural basis for high-affinity fluorophore binding and activation by RNA Mango. *Nat. Chem. Biol.* **13**, 807–813 (2017).
- J. Abramson *et al.*, Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024).
- R. Krishna *et al.*, Generalized biomolecular modeling and design with RoseTTAFold all-atom. *Science* **384**, ead12528 (2024).
- H. M. Berman *et al.*, The protein data bank. *Nucleic Acids Res.* **28**, 235–242 (2000).
- L. F. M. Passalacqua *et al.*, Intricate 3D architecture of a DNA mimic of GFP. *Nature* **618**, 1078–1084 (2023).
- A. He *et al.*, Structure-based investigation of a DNA aptamer targeting PI7K reveals an intricate 3D fold guiding functional optimization. *Proc. Natl. Acad. Sci. U.S.A.* **121**, e2404060121 (2024).
- D. E. Huizenga, J. W. Szostak, A DNA aptamer that binds adenosine and ATP. *Biochemistry* **34**, 656–665 (1995).
- M. Sassanfar, J. W. Szostak, An RNA motif that binds ATP. *Nature* **364**, 550–553 (1993).
- C. H. Lin, D. J. Patel, Structural basis of DNA folding and recognition in an AMP-DNA aptamer complex: Distinct architectures but common recognition motifs for DNA and RNA aptamers complexed to AMP. *Chem. Biol.* **4**, 817–832 (1997).

22. T. Dieckmann, E. Suzuki, G. K. Nakamura, J. Feigon, Solution structure of an ATP-binding RNA aptamer reveals a novel fold. *RNA* **2**, 628–640 (1996).
23. F. Jiang, R. A. Kumar, R. A. Jones, D. J. Patel, Structural basis of RNA folding and recognition in an AMP–RNA aptamer complex. *Nature* **382**, 183–186 (1996).
24. Z. Zhang *et al.*, AMP aptamer programs DNA tile cohesion without canonical base pairing. *J. Am. Chem. Soc.* **145**, 19503–19507 (2023).
25. S. Xie *et al.*, Engineering aptamers with selectively enhanced biostability in the tumor microenvironment. *Angew. Chem. Int. Ed. Engl.* **61**, e202201220 (2022).
26. Y. Ding, J. Liu, Pushing adenosine and ATP SELEX for DNA aptamers with nanomolar affinity. *J. Am. Chem. Soc.* **145**, 7540–7547 (2023).
27. S. Yoshizawa *et al.*, Nuclease resistance of an extraordinarily thermostable mini-hairpin DNA fragment, d(GCGAAGC) and its application to in vitro protein synthesis. *Nucleic Acids Res.* **22**, 2217–2221 (1994).
28. S. S. Wijmenga, B. N. M. van Buuren, The use of NMR methods for conformational studies of nucleic acids. *Prog. Nucl. Magn. Reson. Spectrosc.* **32**, 287–387 (1998).
29. P. Dagenais, G. Desjardins, P. Legault, An integrative NMR-SAXS approach for structural determination of large RNAs defines the substrate-free state of a trans-cleaving *Neurospora Varkud* Satellite ribozyme. *Nucleic Acids Res.* **49**, 11959–11973 (2021).
30. S. L. Lam, L. M. Chi, Use of chemical shifts for structural studies of nucleic acids. *Prog. Nucl. Magn. Reson. Spectrosc.* **56**, 289–310 (2010).
31. J. P. Sheehy, A. R. Davis, B. M. Znosko, Thermodynamic characterization of naturally occurring RNA tetraloops. *RNA* **16**, 417–429 (2010).
32. K. Serec, S. D. Babić, R. Podgornik, S. Tomić, Effect of magnesium ions on the structure of DNA thin films: An infrared spectroscopy study. *Nucleic Acids Res.* **44**, 8456–8464 (2016).
33. A. E. Every, I. M. Russu, Influence of magnesium ions on spontaneous opening of DNA base pairs. *J. Phys. Chem. B* **112**, 7689–7695 (2008).
34. J. H. Tan, A. G. Fraser, Quantifying metabolites using structure-switching aptamers coupled to DNA sequencing. *Nat. Biotechnol.*, 10.1038/s41587-025-02554-7 (2025).
35. K. Yang *et al.*, A functional group-guided approach to aptamers for small molecules. *Science* **380**, 942–948 (2023).
36. F. W. Lowenstein, M. F. Stanton, Serum magnesium levels in the United States, 1971–1974. *J. Am. Coll. Nutr.* **5**, 399–414 (1986).
37. L. S. Green *et al.*, Nuclease-resistant nucleic acid ligands to vascular permeability factor/vascular endothelial growth factor. *Chem. Biol.* **2**, 683–695 (1995).
38. C. P. M. Scheidl, M. Mieczkowski, H. Schindelin, C. Höbartner, Structure and mechanism of the methyltransferase ribozyme MTR1. *Nat. Chem. Biol.* **18**, 547–555 (2022).
39. H. Abou Assi *et al.*, 2'-O-Methylation can increase the abundance and lifetime of alternative RNA conformational states. *Nucleic Acids Res.* **48**, 12365–12379 (2020).
40. O. Alkhamis *et al.*, Exploring the relationship between aptamer binding thermodynamics, affinity, and specificity. *Nucleic Acids Res.* **53**, gkaf219 (2025).
41. T. Hermann, D. J. Patel, Adaptive recognition by nucleic acid aptamers. *Science* **287**, 820–825 (2000).
42. Z. Wang *et al.*, Introducing structure-switching functionality into small-molecule-binding aptamers via nuclease-directed truncation. *Nucleic Acids Res.* **46**, e81 (2018).
43. J. H. de Baaij, J. G. Hoenderop, R. J. Bindels, Magnesium in man: Implications for health and disease. *Physiol. Rev.* **95**, 1–46 (2015).
44. M. J. Abraham *et al.*, GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **1–2**, 19–25 (2015).
45. A. E. Torda, R. M. Cheek, W. F. van Gunsteren, Time-dependent distance restraints in molecular dynamics simulations. *Chem. Phys. Lett.* **157**, 289–294 (1989).
46. K. Hart *et al.*, Optimization of the CHARMM additive force field for DNA: Improved treatment of the BI/BII conformational equilibrium. *J. Chem. Theory Comput.* **8**, 348–362 (2012).
47. J. Huang *et al.*, CHARMM36m: An improved force field for folded and intrinsically disordered proteins. *Nat. Methods* **14**, 71–73 (2017).
48. N. Goga, A. J. Rzepiela, A. H. de Vries, S. J. Marrink, H. J. C. Berendsen, Efficient algorithms for Langevin and DPD dynamics. *J. Chem. Theory Comput.* **8**, 3637–3649 (2012).
49. M. Bernetti, G. Bussi, Pressure control using stochastic cell rescaling. *J. Chem. Phys.* **153**, 114107 (2020).
50. B. Hess, H. Bekker, H. J. C. Berendsen, J. G. E. M. Fraaije, LINC: A linear constraint solver for molecular simulations. *J. Comput. Chem.* **18**, 1463–1472 (1997).
51. D. A. Case *et al.*, AmberTools. *J. Chem. Inf. Model.* **63**, 6183–6191 (2023).
52. W. Humphrey, A. Dalke, K. Schulten, VMD: Visual molecular dynamics. *J. Mol. Graph.* **14**, 33–38 (1996).
53. T. Lu, A comprehensive electron wavefunction analysis toolbox for chemists, Multiwfn. *J. Chem. Phys.* **161**, 082503 (2024).
54. Y. Jiang, L. Wan, P. Guo, Solution NMR structures of ATP-binding DNA aptamer in complex with ATP. Protein Data Bank. <https://doi.org/10.2210/pdb9KTJ/pdb>. Deposited 2 December 2024.